Mechanism design

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Game Theory and Mechanism Design - Lecture 7

(This lecture heavily relies on Narahari (2014) Game Theory and Mechanism Design – chaps. 14 and 15)

Learning objectives

At the end of this lecture (and related tutorial(s)), a student should be able to:

- **Explain** the connection between game theory and mechanism design
- **Provide motivating examples** for such mechanisms
- **Define** social choice functions
- **Assess** whether social choice functions are implementable, or not

Outline

1. Setting the scene
2. Social choice and alternative mechanisms
3. Implementation of social choice function
4. Implementation and Bayesian games

1. **Setting the scene**

From game theory to mechanism design

Mechanism design may be seen as "reverse engineering" game theory

Game Theory

The rules are set, and we want to study the overall outcome

Mechanism Design

We know the overall outcome we want to reach... how to design the game accordingly?

Why is mechanism design appealing?

- it yields decentralised decision-making (instead of centralised optimisation)
- it gives the agent the "illusion of free will"...

It requires what we refer to as a **mechanism designer** (also referred to as social planner)

Cake splitting

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How would you do that?

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- What if you say that one of the kid can cut two pieces, and then the other one can choose which of the pieces they want?

The judgement of Solomon

The Judgment of Solomon (1518-19)
Raffaello Sanzio da Urbino (Raphael) (1483–1520)
National Gallery

2 women recently gave birth to sons, but one of the sons died, leaving the mothers in dispute over the surviving infant. They both claim the surviving infant is theirs!

They are asking for Solomon's wisdom in deciding who gets the baby...

If you were Somolon, **how would you do that?**

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This kind of problem is called **truth elicitation**

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What Solomon did is that he asked one of his men to cut the baby in 2, then each mother will get their share

The two women reacted to this announcement:

- one says "OK, be it, this might be the most optimal and fair outcome" (even though it is clear that no one will have a baby anymore)
- the other says "no - I prefer you give the baby to her than you killing him!"

Solomon gave the baby to the second woman, as she must be the true mother(!)

Mechanism design environment

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- Each agent i has a type θ_i, i.e., most likely some private information they use as a basis to make their decision. The set of types is $\Theta = \prod_i \Theta_i$, and a profile of types is $(\theta_1, \ldots, \theta_n)$

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- There is a common prior for all agents (from Nature), and each agent i can all deduce their posterior beliefs $\phi_i(., \theta_i)$ based on the common prior and the knowledge of their own type θ_i

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- The set of outcomes X, the sets of types for all players Θ_i $(i = 1, \ldots, n)$, the common prior from Nature, the strategy sets and the pay-off functions u_i $(i = 1, \ldots, n)$ are common knowledge. **Only the type of the agents is private knowledge**, i.e., only agent i knows they are of type θ_i.

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That could look like a Bayesian game setup... no?

2. **Social choice and alternative mechanisms**

(Let us reflect on what we really want to do here!)

Social choice function

We have mentioned that all these agents (/players) are somewhat to make a *collective* decision

Definition (Social choice function)

Considering a set of agents $N = \{1, \ldots, n\}$, with sets of types $\Theta_1, \ldots, \Theta_n$. Given a set of outcomes X, a *social choice function* is a mapping of the form

$$f : \Theta_1 \times \ldots, \times \Theta_n \to X .$$

It means that f assigns a given outcome $x \in X$ to each possible combination of types for all agents involved (where a combination is given by a profile of types $(\theta_1, \ldots, \theta_n)$).

The outcome corresponding to a given type profile $(\theta_1, \ldots, \theta_n)$ is called the *social choice* (or *collective choice*) for that type profile.

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Can you describe mechanisms and social choice function from real-world problems?

Two types of mechanisms

Sometimes, the point is directly get the agents to tell about their types...

Definition (Direct mechanisms)

Given a social choice function $f : \Theta_1 \times \ldots, \times \Theta_n \to X$, a *direct mechanism* is defined as the tuple $\langle \Theta_1, \ldots, \Theta_n, f \rangle$.

Two types of mechanisms

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Definition (Direct mechanisms)

Given a social choice function $f : \Theta_1 \times \ldots, \times \Theta_n \rightarrow X$, a *direct mechanism* is defined as the tuple $\langle \Theta_1, \ldots, \Theta_n, f \rangle$.

And, sometimes, they will just choose a strategy (/action) based on their type...

Definition (Indirect mechanisms)

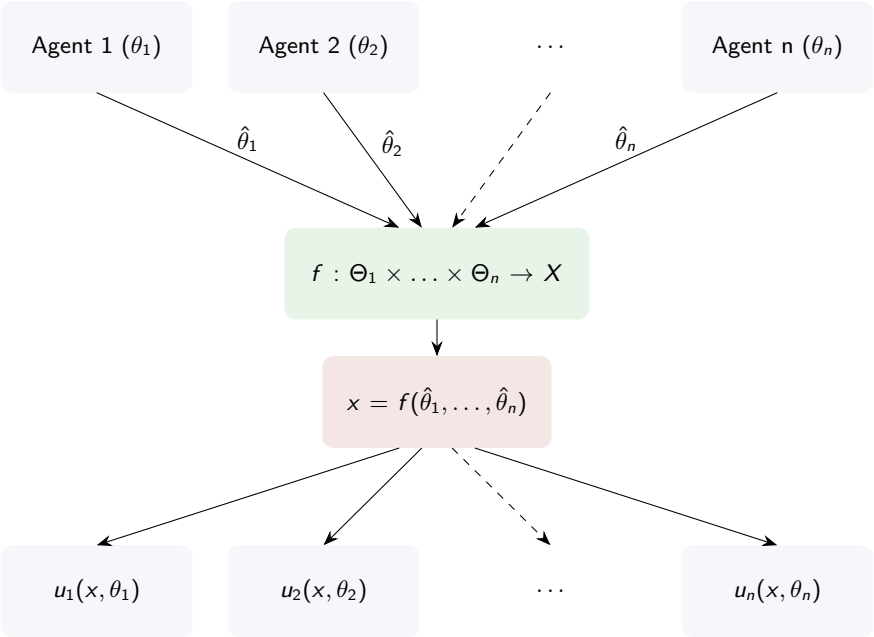
Instead of relying on types, an *indirect mechanism* considers the connection between the strategies used by the agents, given their type, and the outcomes. I.e., an indirect mechanism is defined as a tuple $\langle S_1, \ldots, S_n, g \rangle$, where S_i is the strategy set of agent i and $g : S_1 \times \ldots, \times S_n \rightarrow X$. It matches is potential strategy (/action) profile to a given outcome.

A direct mechanism actually is a special case of an indirect mechanism, since then the strategy (/action) is about to reveal one's type, or not. Hence,

- $S_i = \Theta_i, \ \forall i$
- $f = g$

Therefore, we mostly write the following results for indirect mechanisms...

Illustration of the overall process (direct mechanism)



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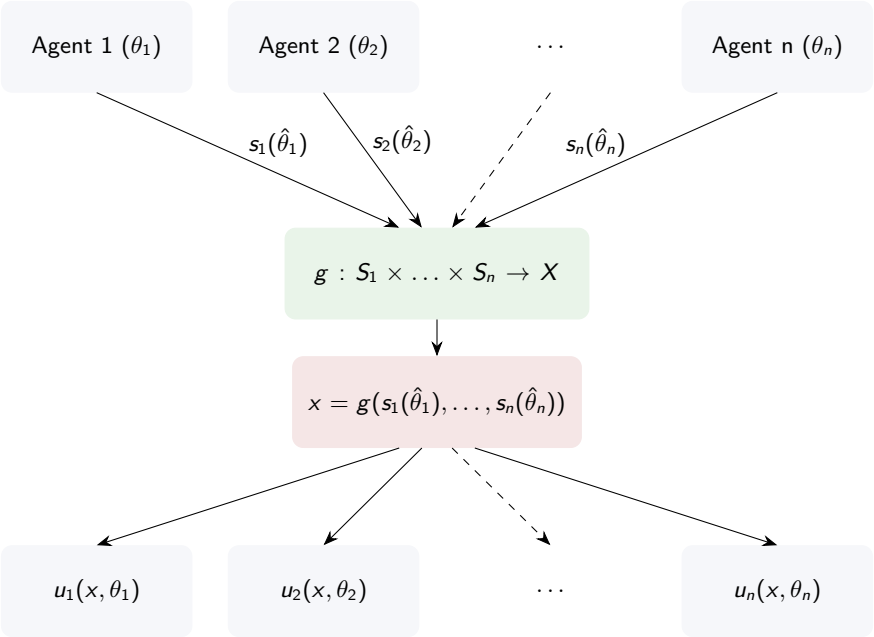

An example - supplier selection (1)

Apple iPhone (a_1)

Samsung A13/Samsung S24 (b_2 and a_2)

Supplier selection problem

Imperial College (mechanism designer) wants to select a supplier for phones for employees (out of $N = \{1, 2\}$)

- Player 1 (supplier) can procure a single phone, $\Theta_1 = \{a_1\}$
- Player 2 (supplier) has both a high-quality (a_2) and low-quality (b_2) phone on offer, $\Theta_2 = \{a_2, b_2\}$

This means that $\Theta = \Theta_1 \times \Theta_2 = \{(a_1, a_2), (a_1, b_2)\}$.

There are 3 potential outcomes $X = \{x, y, z\}$:

- Outcome x – all phones supplied by player 1
- Outcome y – half of the phones supplied by player 1, the other half by player 2
- Outcome z – all phones supplied by player 2

The issue is that player 1 is the historically preferred provider, but a_2 (player 2) definitely is a superior technology that is attractive...

An example - supplier selection (2)

Apple iPhone(a_1)

Samsung A13/Samsung S24 (b_2 and a_2)

Supplier selection problem

Consider the following **pay-offs**:

- Player 1 (type a_1)

$$u_1(x, a_1) = 100, \;\; u_1(y, a_1) = 50, \;\; u_1(z, a_1) = 0$$

- Player 2 (type a_2)

$$u_2(x, a_2) = 0, \;\; u_2(y, a_2) = 50, \;\; u_2(z, a_2) = 100$$

- Player 2 (type b_2)

$$u_2(x, b_2) = 0, \;\; u_2(y, b_2) = 50, \;\; u_2(z, b_2) = 25$$

An example **social choice function** (remember that $f : \Theta \to X$) is

$$f(a_1, a_2) = y, \qquad f(a_1, b_2) = x$$

Do you see **other potential social choice functions**?

An example - supplier selection (3)

Apple iPhone(a_1)

Samsung A13/Samsung S24 (b_2 and a_2)

Supplier selection problem

The list of all potential social choices (3^2 of them):

$f(a_1, a_2) = x$	$f(a_1, b_2) = x$
$f(a_1, a_2) = x$	$f(a_1, b_2) = y$
$f(a_1, a_2) = x$	$f(a_1, b_2) = z$
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$f(a_1, a_2) = z$	$f(a_1, b_2) = y$
$f(a_1, a_2) = z$	$f(a_1, b_2) = z$

Do they all make sense to you?

3. **Implementing the social choice function**

Analysing the supplier selection problem

Supplier selection problem

Pay-offs:

$$u_1(x, a_1) = 100, u_1(y, a_2) = 50, \quad u_1(z, a_1) = 0$$

$$u_2(x, a_2) = 0, \quad u_2(y, a_2) = 50, \; u_2(z, a_2) = 100$$

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Outcomes:

- x – all phones supplied by player 1
- y – half of the phones supplied by player 1, the other half by player 2
- z – all phones supplied by player 2

The mechanism designer would like to *implement* the social choice $f(a_1, a_2) = y$, $f(a_1, b_2) = x$, and hence wants the agents to reveal their type.

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Focus on Player 1:

The type of Player 1 is common knowledge – hence, it is already revealed(!)

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The type of Player 1 is common knowledge – hence, it is already revealed(!)

Focus on Player 2:

- if $\theta_2 = a_2$, with $f(a_1, a_2) = y, \; f(a_1, b_2) = x$, and since $u_2(y, a_2) > u_2(x, a_2)$, revealing their true type is a best response for Player 2
- if $\theta_2 = b_2$, since $u_2(y, b_2) > u_2(x, b_2)$, Player 2 sees that it would best for them to lie about their type and say that their type is a_2...(!)

Analysing the supplier selection problem

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We say that **this social choice function is not implementable**.

Analysing the supplier selection problem

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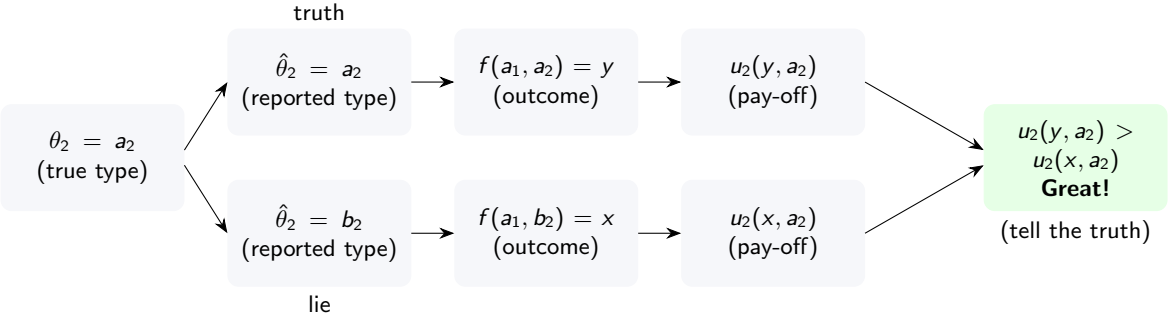
Case of $\theta_2 = a_2$



Analysing the supplier selection problem

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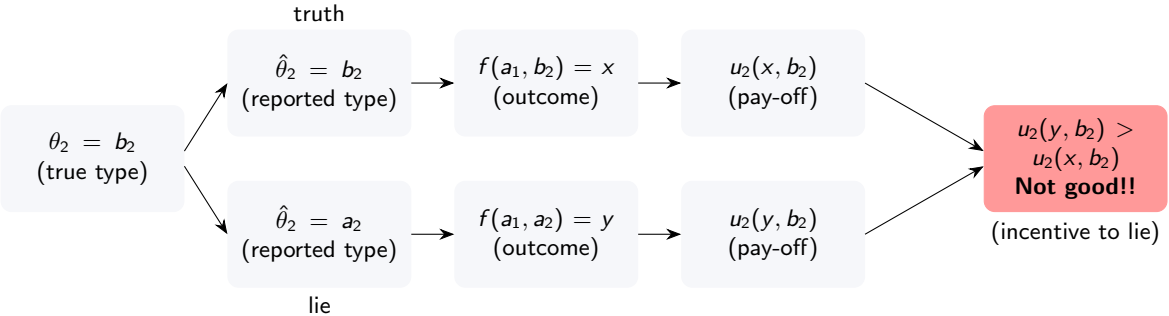

Not all social choice functions are implementable

Apple iPhone(a_1)

Samsung A13/Samsung S24 (b_2 and a_2)

In some case, the overall objective of the mechanism designer is aligned with the agents' interests, sometimes they are not.

Supplier selection problem

Implementable (👍) and non-implementable (👎) social choice functions:

$f(a_1, a_2) = x$	$f(a_1, b_2) = x$	👍
$f(a_1, a_2) = x$	$f(a_1, b_2) = y$	👎
$f(a_1, a_2) = x$	$f(a_1, b_2) = z$	👎
$f(a_1, a_2) = y$	$f(a_1, b_2) = x$	👎
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$f(a_1, a_2) = z$	$f(a_1, b_2) = y$	👍
$f(a_1, a_2) = z$	$f(a_1, b_2) = z$	👍

We can say that these social choice functions are implementable by a direct mechanism (it is not always the case...)

4. **Implementation and Bayesian games**

Bayesian games induced by a mechanism

As for our setup, we have

- N agents, $i \in \{1, \ldots, n\}$
- sets of types Θ_i, $i = 1, \ldots, n$
- a common prior, as well as posterior beliefs $\phi_i(.|\theta_i)$, $i = 1, \ldots, n$
- a set of outcomes $X = \{x_1, \ldots, x_m\}$
- pay-off (/utility) functions $u_i(x, \theta_i)$ that are type-dependent (where $x \in X$ is one of the outcomes and $\theta_i \in \Theta_i$ is one of their potential types)

Based on these input conditions...

Definition (Mechanism-induced Bayesian game)

Denote an indirect mechanism $\mathcal{M} = \langle (S_i), g \rangle$. Based on the above conditions, $\mathcal{M}$ induces a Bayesian game $\Gamma = \langle N, (\theta_i), (S_i), (\phi_i), (u_i) \rangle$ among the players, where

$$u_i(\theta_1, \ldots, \theta_n, s_1, \ldots, s_n) \; = \; u_i(g(s_1, \ldots, s_n), \theta_i)$$

Implementation of a social choice function

The mechanism designer is not a *dictator*..

They have a view on what they would like the outcome to be, but they would like for it to happen naturally through the game setup induced by their mechanism.

Definition (Implementation of a social choice function)

A mechanism $\mathcal{M} = \langle (S_i), g \rangle$, with $g : \prod_i S_i \to X$ *implements the social choice function* f if there exists a pure strategy equilibrium $s^* = (s_1^*, \ldots, s_n^*)$ of the Bayesian game Γ induced by $\mathcal{M}$ such that

$$g(s_1^*(\theta_1), \ldots, s_n^*(\theta_n))) = f(\theta_1, \ldots, \theta_n), \quad \forall (\theta_1, \ldots, \theta_n) \in \Theta$$

This is a very powerful idea, since:

- the mechanism designer is not a dictator, but will find a way to get certain outcomes out of the interaction between players
- the players get there by themselves, and since it leads to an equilibrium, it is an outcome that makes sense to them!

Implementation and strategies

Now that we know we have a Bayesian game, we can think of *different types* of equilibria...

Definition (Implementation in dominant strategy equilibrium)

A mechanism $\mathcal{M} = \langle (S_i), g \rangle$ is said to *implement* the social choice function f in *dominant strategy equilibrium*, if there is a weakly dominant equilibrium $s^* = (s_1^*, \ldots, s_n^*)$ of the Bayesian game Γ induced by $\mathcal{M}$ such that

$$g(s_1^*(\theta_1), \ldots, s_n^*(\theta_n)) = f(\theta_1, \ldots, \theta_n), \quad \forall (\theta_1, \ldots, \theta_n) \in \Theta$$

This result is written for the indirect mechanism case (since more general). The same can be said for a direct mechanism.

Implementation and strategies

Now that we know we have a Bayesian game, we can think of *different types* of equilibria...

Definition (Implementation in dominant strategy equilibrium)

A mechanism $\mathcal{M} = \langle (S_i), g \rangle$ is said to *implement* the social choice function f in *dominant strategy equilibrium*, if there is a weakly dominant equilibrium $s^* = (s_1^*, \ldots, s_n^*)$ of the Bayesian game Γ induced by $\mathcal{M}$ such that

$$g(s_1^*(\theta_1), \ldots, s_n^*(\theta_n)) = f(\theta_1, \ldots, \theta_n), \quad \forall (\theta_1, \ldots, \theta_n) \in \Theta$$

This result is written for the indirect mechanism case (since more general). The same can be said for a direct mechanism.

Definition (Implementation in Bayesian Nash equilibrium)

A mechanism $\mathcal{M} = \langle (S_i), g \rangle$ is said to *implement* the social choice function f in *Bayesian Nash equilibrium*, if there is a pure strategy equilibrium $s^* = (s_1^*, \ldots, s_n^*)$ of the Bayesian game Γ induced by $\mathcal{M}$ such that

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Implementation and strategies

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A mechanism that is implementable in some type of equilibrium is called **incentive-compatible**: none of the agents involved has an incentive *not to* report, or act upon, their true type(!)

Wrapping up

- Mechanism design is **reverse engineering** game theory
- Agents have their **types** and related **pay-offs**, while the mechanism designer has their **social choice**
- A social choice (see it as the target outcome) is **not necessarily implementable!**
- Implementing social choice functions is a core issue for many **multi-agent engineering and social systems**

Further readings (/lecture notes for this topic):

Narahari Y (2014) *Game Theory and Mechanism Design* - Chaps. 14-15 (available on Blackboard Learn)

Thanks for your attention!